The $d = 4$ fiber: a π-adic engine, and even-$|J^*|$ reduced to a mod-2 identity

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Setup. $\lambda \vdash n = 2m$; $G_\lambda(i) = \langle s_\lambda, \psi^m \rangle$, $\psi = h_2 + ie_2$, is the fiber at $\zeta_4 = i$ of $G_\lambda(q) = \sum_{T \in \mathrm{SYT}(\lambda)} q^{s(T)}$. Write $\pi = 1 + i$, so $\mathbb{Z}[i]/(\pi) \cong \mathbb{F}_2$, $v_\pi(N) = 2v_2(N)$ for $N \in \mathbb{Z}$, $\pi^2 = 2i$. With $h_2 = \frac{1}{2}(p_1^2 + p_2)$, $e_2 = \frac{1}{2}(p_1^2 - p_2)$ one has $p_1^2 = p_2 + 2e_2$; set $\chi_k := \chi^\lambda(2^k 1^{2m-2k}) = \langle s_\lambda, p_2^k p_1^{2m-2k} \rangle$, $f^\lambda = \chi_0$.

1 Reformulation

Lemma 1. *$\psi = p_2 + \pi e_2$, hence $G_\lambda(i) = \Psi(\pi)$ where $\Psi(Y) = \sum_{r=0}^m \binom{m}{r} R_r Y^r$, $R_r = \langle s_\lambda, p_2^{m-r} e_2^r \rangle \in \mathbb{Z}$, $R_0 = \chi^\lambda(2^m)$.*

Proof. $h_2 = p_2 + e_2$ gives $\psi = p_2 + (1+i)e_2$; expand ψ^m and pair with s_λ. ☐

The term valuations are $\mathrm{val}(r) = r + 2v_2(\binom{m}{r} R_r) \geq 0$, a π-adic Newton polygon for G (minimum locus $= J^*$). Also $p_1^2 - ip_2 = -i\pi(p_2 + \pi e_2)$, so with $\Phi(z) := \langle s_\lambda, (p_1^2 + zp_2)^m \rangle = \sum_k \binom{m}{k} \chi_k z^k$, $G_\lambda(i) = (\frac{1+i}{2})^m \Phi(-i)$ and $G_\lambda(i) = 0 \iff (z^2 + 1) \mid \Phi(z)$ in $\mathbb{Z}[z]$.

2 A one-line congruence

Proposition 2. *$G_\lambda(i) \equiv \chi^\lambda(2^m) \pmod{\pi}$.*

Proof. In $\Psi(\pi) = \sum_r \binom{m}{r} R_r \pi^r$ every $r \geq 1$ term is $\equiv 0 \pmod{\pi}$, leaving $R_0 = \chi^\lambda(2^m)$. ☐

Lemma 3. *$\chi_k \equiv f^\lambda \pmod 2$ for all k.*

Proof. $p_1^2 \equiv p_2 \pmod 2$, so $p_2^k p_1^{2m-2k} \equiv p_1^{2m} \pmod 2$; pair with the integral s_λ. ☐

Corollary 4. *$\chi^\lambda(2^m) \equiv f^\lambda \pmod 2$; thus f^λ odd $\Rightarrow G_\lambda(i) \not\equiv 0 \pmod{\pi} \Rightarrow G_\lambda(i) \neq 0$, and every tie ($|J^*| > 1$) has f^λ even.*

3 The 2-adic engine

Theorem 5. $\Phi(z) = \displaystyle\sum_{r=0}^m \binom{m}{r} 2^r R_r \, (1+z)^{m-r}$.

Proof. $p_1^2 + zp_2 = (1+z)p_2 + 2e_2$; expand and pair with s_λ. ☐

Corollary 6. *$\Phi(z) \equiv \chi^\lambda(2^m)(1+z)^m \pmod 2$. If $\chi^\lambda(2^m)$ is odd, the coarse ($z = -i$, a unit) Newton locus $B^* = \arg\min_k v_2(\binom{m}{k}\chi_k)$ equals the set of binary submasks of m (Lucas), so $|B^*| = 2^{s_2(m)}$.*

4 The box, and the reduction

For a polynomial evaluated at a unit, the minimal-valuation locus is an affine 2-adic box $j_0 + \{\sum_{a\in S} 2^a\}$ iff, after dividing by 2^e ($e = \min v_2$ of coefficients), the mod-2 reduction is $z^{j_0}(1+z)^g$ (over $\mathbb{F}_2$, $(1+z)^g = \prod_{a\in S}(1+z^{2^a})$ has exactly that support). The sharp polygon J^* lies in one parity class (Fact 4.1); restricting to it and absorbing the parity tilt 2^s gives a scaled polynomial whose bottom-mod-2 part is the analog.

Conjecture 7 (Box)**.** Coarse: $\Phi(z)/2^e \equiv z^{j_0}(1+z)^g \pmod 2$. Sharp: the parity-restricted tilt-scaled polynomial is $\equiv y^{l_0}(1+y)^{g'} \pmod 2$. Hence B^*, J^* are affine 2-adic boxes, $|J^*|$ is a power of two, and even on a tie.

The engine (Cor. 7) proves the $r = 0$ layer of Theorem 6. When $\chi^\lambda(2^m)$ is even (exactly the tie case) that layer vanishes mod 2 and the next layer involves e_2 mod 2, where $p_1^2 \equiv p_2$ no longer collapses the two generators; a uniform proof needs the statement mod π^k. Verified exhaustively for all $\lambda \vdash 2m$, $m \leq 12$: coarse 4114/4114, sharp 1624/1624 ties; $|B^*|, |J^*| \in \{1, 2, 4, 8\}$.

Honest scope. This does not close non-vanishing: on a tie the leading π-coefficient cancels and $v_\pi(G)$ is set by an interleaved cancellation tower of unbounded depth ($d = v_\pi(G) - \mu$ ranges 1–22, $m \leq 12$). The unique total vanisher is $(2, 2)$, where $\Phi = 2(1 + z^2)$.